

Course syllabus for

Academic year 2022/2023 >

TIF091 - Experimental physics 2 - basis

Experimentell fysik 2 - bas

Course syllabus adopted 2021-02-26 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

4,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Engineering Physics

Department: 16 - PHYSICS

Teaching language: Swedish

Application code: 57137

Open for exchange students: No

Maximum participants: 100

Only students with the course round in the programme overview

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0119 Laboratory 3,0 c Grading: UG	1,5 c	1,5 c					
0219 Project 1,5 c Grading: TH			1,5 c				

In programs

[TKTFY ENGINEERING PHYSICS, Year 3 \(compulsory elective\)](#)

Examiner:

Jan Swenson



[Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Experimental physics 1

Aim

Introduces new forms of experimental work where the knowledge from earlier theoretical courses are used.

Learning outcomes (after completion of the course the student should be able to)

Develop the ability to do good experimental physics.

Learn to formulate problems oneself.

Build up experimental set ups.

Make accurate measurements with advanced equipment.

Content

This is a laboratory course with a main goal of developing experimental skills. Students work with three problems throughout the school year in the areas of Optics, Thermodynamics, Quantum Physics. Each problem consists of two parts; one that is mandatory to work with to get a passing grade and a second part where students address a more complex aspect of the problem based on their own ideas. A technical report based on the work carried out has to be submitted for grading. Students work in pairs. A 15 minutes interview with the instructor is scheduled a week in advance of the lab time for presentation of the main ideas of how the students intend to solve the problem.

Organisation

Students work in groups of two with 3 different problems covering different areas of physics. To carry out the experimental work students have access to advanced equipment over a 3-4 day period. Projects are allocated well in advance of the lab time. Two of the problems are included in Laboratory during the fall term, whereas the third one is included in the Project.

Literature

Relevant manuals. Additional reading material on the web page of the course. Literature for the theoretical courses. For details, see the home page.

Examination including compulsory elements

The technical report has to be handed in at latest one month after the lab time. Each problem contains two parts: a basic project and an extra project. The former must be solved for a passing grade (3). For a higher grade (4 or 5) the student needs to work with the extra part. To pass the laboratory part during the fall term the student needs to get a pass on the preliminary interview and to solve the two problems included. The grade on the project is the grade on the whole course. Grading is based on the three submitted technical reports. A passing grade on the course requires passing grades on both Laboratory and Project.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.